

1. Administrative & Study Identification

Full Study Title: A Phase III, Open-Label, Randomized Study of Atezolizumab With Lenvatinib or Sorafenib Versus Lenvatinib or Sorafenib Alone in Hepatocellular Carcinoma Previously Treated With Atezolizumab and Bevacizumab **Short Title/Acronym:** IMbrave251 **Protocol Number:** M042541 / 2023-503229-21-00 **NCT Number:** NCT04770896 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Atezolizumab, Lenvatinib, Sorafenib **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of atezolizumab plus either lenvatinib or sorafenib versus lenvatinib or sorafenib alone in participants with locally advanced or metastatic Hepatocellular Carcinoma (HCC) who have progressed on prior systemic treatment with the atezolizumab plus bevacizumab combination. * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Confirmed Objective Response Rate (ORR), and Time to Progression (TTP).

2.2 Background & Rationale Hepatocellular carcinoma (HCC) is the most common type of liver cancer. Standard first-line medicine for inoperable HCC includes atezolizumab and bevacizumab, which have become the standard of care. However, when these first treatments stop working and the disease progresses, there is currently no definitive standard second-line therapy regimen with strong evidence supporting its use. The IMbrave251 study addresses this clinical gap by evaluating whether continuing immune checkpoint inhibition (atezolizumab) in combination with a multi-kinase inhibitor (lenvatinib or sorafenib) provides superior survival outcomes compared to switching to a multi-kinase inhibitor alone.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Phase Criteria:** Trial must be in PHASE3 (criteria_type: PHASE_REQUIREMENT, phase_id: PHASE3) * **Control Type:** Active Comparator (Lenvatinib or Sorafenib alone) * **Blinding:** Open-label * **Randomization:** Randomized (1:1 allocation)

3.2 Planned Enrollment & Duration * Target N\$: 557 participants
*** Number of Sites: Multicenter (Global) * Estimated Study Start:**
26-Apr-2021 * Estimated Primary Completion: 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Locally advanced or metastatic and/or unresectable HCC with diagnosis confirmed by histology/cytology or clinically by American Association for the Study of Liver Diseases (AASLD) criteria in cirrhotic patients. 2. Disease progression following prior atezolizumab plus bevacizumab combination treatment for HCC, for at least 4 consecutive treatment cycles, and 2 subsequent tumor assessments (with at least 1 assessment showing SD, PR, or CR). 3. At least one measurable target lesion per RECIST v1.1 that has not been previously treated with local therapy or, if the target lesion is within the field of previous local therapy, has subsequently progressed in accordance with RECIST v1.1. 4. Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0 or 1 within 7 days prior to randomization. 5. Child-Pugh class A within 7 days prior to randomization. 6. Adequate hematologic and end-organ function.

4.2 Exclusion Criteria 1. Symptomatic, untreated, or actively progressing central nervous system (CNS) metastases. 2. History of leptomeningeal disease. 3. History of hepatic encephalopathy in the preceding 6 months, unresponsive to therapy within 3 days. 4. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 5. History of malignancy other than HCC within 5 years prior to screening, with the exception of malignancies with a negligible risk of metastasis or death.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Participants are randomized to receive either atezolizumab + (lenvatinib or sorafenib) OR (lenvatinib or sorafenib) alone. Lenvatinib is administered orally once daily in 21-day cycles: 8 mg daily for baseline body weight < 60 kg, or 12 mg daily for baseline body weight $\geq$ 60 kg. *** Specific Drug Details (Sorafenib):** *** Preferred Name:** Sorafenib *** Trade Name:** Nexavar *** ATC Code:** L01EX02 *** EMA URL:** [Nexavar EPAR Product Information](#) *** Route of Administration:** Intravenous (Atezolizumab) and Oral (Lenvatinib, Sorafenib). *** Duration of Treatment:** Treatment will continue until unacceptable toxicity or loss of clinical benefit as determined by the investigator.

5.2 Concomitant & Prohibited Therapy * Permitted: Standard supportive care for underlying cirrhosis and symptom management. *** Prohibited:** Any concurrent systemic anti-cancer therapies or investigational agents not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical/Cancer History, Tumor Imaging, Eligibility Labs
Baseline	Day 0	Randomization, First Dose Administration	
Interim	Every 21 days	Study Treatment Cycles, Safety Labs, Efficacy Assessment (RECIST v1.1)	End of Study Variable
Disease Progression, Unacceptable Toxicity, Exit Interview			

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Overall Survival (OS), defined as the time from randomization into the study to death from any cause. * **Measurement Method:** Investigator and survival follow-up assessments.

7.2 Secondary & Exploratory Endpoints * **Progression-Free Survival (PFS):** Time from randomization to the first occurrence of disease progression or death. * **Confirmed Objective Response Rate (ORR):** Proportion of patients with a best response of CR or PR. * **Time to Progression (TTP):** Time from randomization to first occurrence of disease progression.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via onsite visits and clinical laboratory assessments prior to each treatment cycle.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor/Regulatory Authorities.
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to evaluate ongoing safety and efficacy profiles.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy endpoints (OS, PFS). Safety Set for all participants receiving at least one dose of the study drug.
 - **Sample Size Justification:** $N = 557$ participants, randomized 1:1, powered to detect a statistically significant improvement in Overall Survival (OS) between the two treatment arms.
 - **Handling of Missing Data:** Censoring rules applied for survival and progression endpoints per standard RECIST v1.1 and oncology statistical guidelines.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and centralized remote monitoring by the Sponsor to ensure protocol adherence and patient safety.
 - **Audit Trail:** eCRF entry locking, central radiological review, and source data verification.
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11. Study Identifiers & Governance

- **Primary ID:** M042541
 - **Registry Number:** NCT04770896
 - **Posting ID:** 403576409948405956
 - **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche)
 - **Study Title:** IMbrave251 / A Study of Atezolizumab With Lenvatinib or Sorafenib Versus Lenvatinib or Sorafenib Alone in Hepatocellular Carcinoma Previously Treated With Atezolizumab and Bevacizumab
 - **Official Regulatory Title:** A Phase III, Open-Label, Randomized Study of Atezolizumab With Lenvatinib or Sorafenib Versus Lenvatinib or Sorafenib Alone in Hepatocellular Carcinoma Previously Treated With Atezolizumab and Bevacizumab
 - **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)
 - **Source Level:** 5
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12. Clinical Framework (Hepatocellular Carcinoma Specific)

- **Primary Condition / Preferred Name:** {{Unresectable Hepatocellular Carcinoma (HCC)}}
- **Preferred UMLS Name:** {{Hepatocellular Carcinoma, Drug Therapy}}
- **Condition Definition:** {{A hepatocellular carcinoma that is not amenable to surgical resection.}}
- **Semantic Type:** {{Neoplastic Process}}
- **MeSH Code:** {{C06.301.760.300}}
- **SNOMED ID:** {{447833005}}
- **SNOMED Hierarchy:** {{372138002|Hepatocellular carcinoma|Hepatocellular carcinoma (disorder)}}
- **Diagnosis Threshold:** {{Diagnosis confirmed by histology/cytology or clinically by AASLD criteria}}
- **Medical Methodology:** {{Sequential systemic therapy combining Immune Checkpoint Inhibitor (ICI) with Tyrosine Kinase Inhibitors (TKI)}}

- **Optimization Target:** {{Overall Survival (OS) and Progression-Free Survival (PFS) improvement}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Second-line advanced HCC management post-Atezolizumab/Bevacizumab progression}}
 - **Education Protocol (HCP):** {{Protocol adherence, AE management, and RECIST v1.1 tumor assessment training}}
 - **Education Protocol (Patient):** {{Onsite training on oral TKI adherence and immune-related adverse event (irAE) self-monitoring}}
 - **Quality Audit Mechanism:** {{Routine site monitoring and independent central radiological review facility}}
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14. Observation & Follow-Up Schedule

- **Total Duration:** {{Continuous until unacceptable toxicity or loss of clinical benefit}}
 - **Onsite Visit Cadence:** {{Every 21-day cycle}}
 - **Remote Monitoring Frequency:** {{As needed for survival tracking and safety follow-up}}
 - **Checklist Review Interval:** {{Coincides with periodic tumor imaging schedules}}
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				